Thermo________

relating to temperature

heat

_chemistry

Study of matter

Thermochemistry

Chapter 9

Chemistry: Atoms First 2nd Ed.

Louis Pascal Riel

UConn Avery Point

Spring 2026

1127 Housekeeping Apr 28th Quiz 7 Topics

- HW see due dates →
  A12 & A13 **BOTH** Due Today 11:59 PM!!
  A14 Due Thurs Apr 30th 11:59 PM!!
  A15 & A16 Due Tues May 5th 11:59 PM!!
  A17 Due Tuesday May 5th 11:59 PM! Will cover a missed assignment or be extra credit
- Extra Credit – 4.5 pt Exam Review Game
  - Of your 5 questions, have at least one come from Exam 1, 2, 3, and quiz 7
  - Problems Due Today Apr 28th 4:00 PM
- EC – Discussion Board by Monday – 2 pt for posting
  - Due Mon May 4th 4:00 PM
- Quiz 7 – Thurs Apr 30th

My Final Exam Schedule > Spring 2026 > University of Connecticut

Class	Exam Date	Exam Time	Exam Room
CHEM 1128Q-660 (12412)	5/5/2026, Tuesday	8:00AM - 10:00AM	ACD 308
CHEM 1127Q-660 (12409)	5/7/2026, Thursday	8:00AM - 10:00AM	ACD 211
CHEM 2444-660 (12416)	5/7/2026, Thursday	10:30AM - 12:30PM	ACD 207

Extra Credit Bonanza Spring 2026

- Exam Final Review Game b/c of Quiz 7 – see Previous Slide
- Final Exam Extra Credit (HuskyCT Discussion Board) **2 pts**:
    - Complete by Monday May 4th 4:00 pm

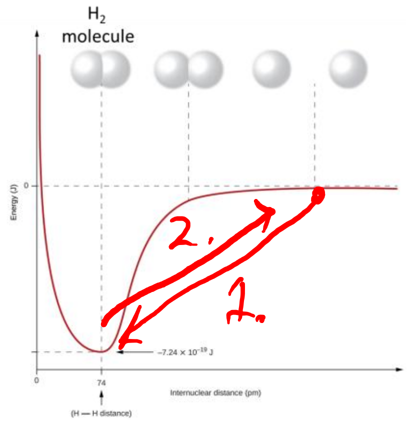
Current Response Rate: 7/16, 44%

- SET
    - Now Available Extra credit: My final check of SET response rates will be **Thursday April 30th 4:00 pm**
    - Once 50% response rate is met, **+ 2 pts** on final & for every 10% threshold more, **+ 0.5 pts**
    - 55% - +2 60% - +2.5 70% - +3 87% - +3.5 100% - **4.5** pts
    - I value this feedback! I incorporated recordings and AktivCHEM HW in part do to feedback from the end of the fall semester.
- Pedagogical Improvement: **2 pts max** extra credit for helping my lectures improve, by submitting (via HuskyCT Discussion Board) a
    - I want to hear about classes you have had that incorporate game/competitive aspects.
    - Diagram/drawing/art/image that you think would improve my lecture.
    - Mnemonics or ways you remember some of the material or real-world example of some chemistry we have discussed.
    - Youtube video or Web link (plus a short written explanation for why you think it was helpful).
    - Complete by Monday May 4th 4:00 pm

HW for the following material will be titled
A16: Chapter 9 First Portion

Thermochemistry - Introduction

- Many chemical reactions involve changes in both **matter** and **energy**
  - i.e. lighting/combusting a match



1. **Making** a chemical bond release energy

One of these two **will likely be a question on your quiz 7 & Final**

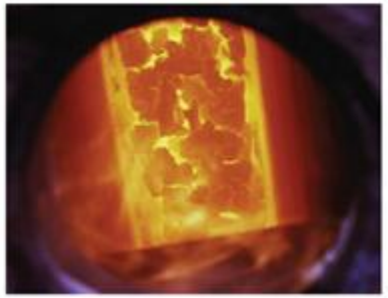
2. **Breaking** a chemical bond requires energy

Think magnets being broken apart

absorbs

- If a reaction involves multiple bonds being made and broken, must account for **all bonds being made and/or broken**.

Energy

- Chemical changes involve energy changes.

Energy released or absorbed?

- **Energy** is the capacity to affect change by supplying **heat** or doing **work**
  - **Work** is causing matter to move against an opposing force.
  - **Heat** is the transfer of thermal energy between an object at high temperature to an object at low temperature.

- **Thermochemistry** – area of science that deals with the amount of **heat absorbed** or **released** through chemical or physical changes.

required,

Two Types of Energy

1. **Potential Energy** – is the energy due to an object's relative **position** (i.e. gravity), or **composition**
   - A Dam holding back water
   - TNT

2. **Kinetic Energy** – is the energy an object possesses because of its motion or **movement**
   - Rocket moving
   - Falling water
   - Thermal Energy

Water has both
PE & KE
in these pictures

High PE

KE

Low PE

ICE Energy

39:30

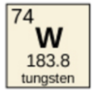
- 45 s

- Kinetic Energy is energy of movement
- KE = ½ m * v^2
  - m = mass, v = velocity (speed)

The Law of Conservation of Energy

- Energy can neither be created nor destroyed
  - Energy can be converted from one form to another.

- This is one version/aspect of the 1st Law of Thermodynamics

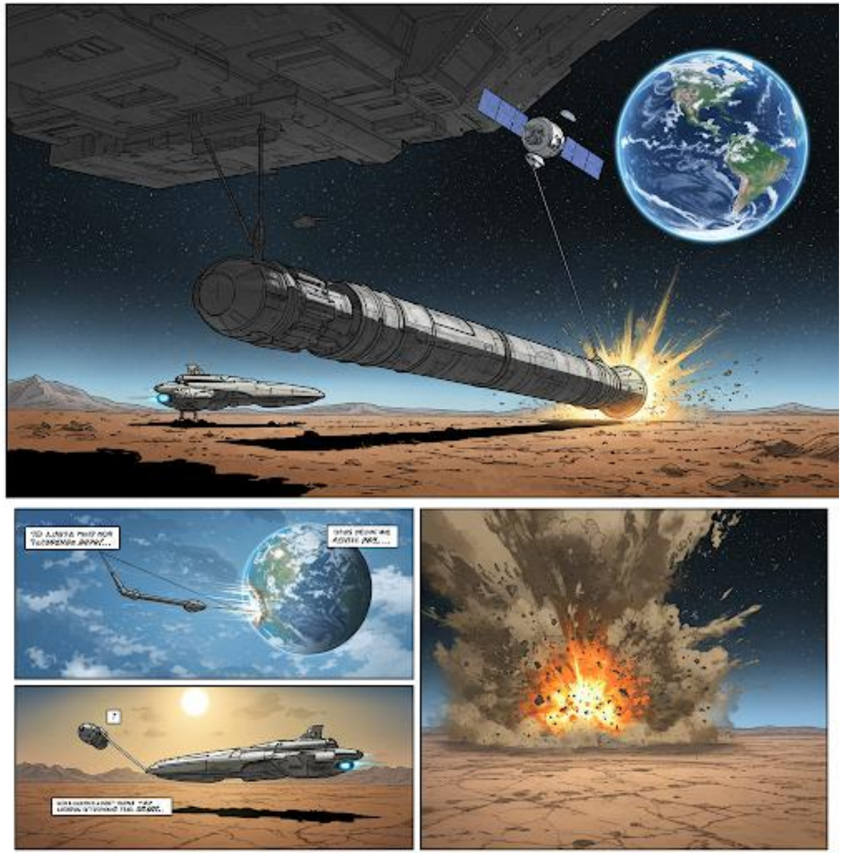


- Rods from god – theoretical ballistic weapon system whereby W rods are stored in orbit on satellites
  - Consider the conversion of energy involved in this
    - Space Shuttle w/ Rod, Orbiting Satellite w/ Rod, Rod Plummeting, Impact
  - I want someone to make me a 4-panel cartoon of this, as Gemini has not

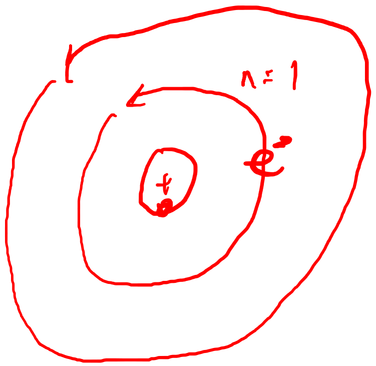
Fuel tanks ↑↑ P.E. KE High P.E. KE

Rods from God, Thx Gemini

- My Ask: Hi! I hope you can create me a 4 panel cartoon pertaining to the theoretical space-based ballistic weapon, a rod from god. This is where a large heavy rod is dropped from orbit and causes significant damage with no radioactive fallout. I want the first panel to be of a spaceship still on land with the rod in tow. The second panel to be of the rod attached to a satellite orbiting earth, the third to be the rod released and plummeting back towards the earth's surface, and the fourth being the cloud that is beginning to start right after it impacts the earth.



Energy can be converted from one form to another.

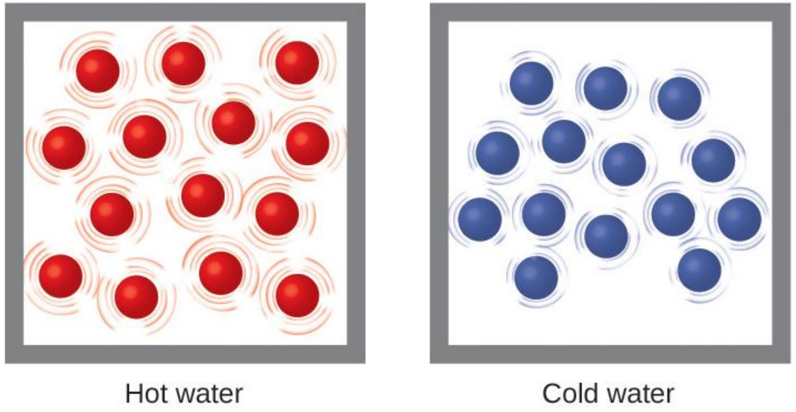


- Changes include
  - Absorbing or releasing heat
  - Conversion into or from light
    - Using $c = \lambda \nu$ & $E = h \nu$
  - Doing work
  - Chemical bond forming to break a different bond.

- In chemical reactions, energy and mass are both conserved.

Higher Temperature = High Thermal Energy

- Thermal energy is a type of **kinetic energy** (KE) associated with the random motion of atoms and molecules
- **Temperature** is a **quantitative** measure of ‘hot’ or ‘cold’
  - An object that is moving/vibrating **quickly** has a higher average KE and is **hot**



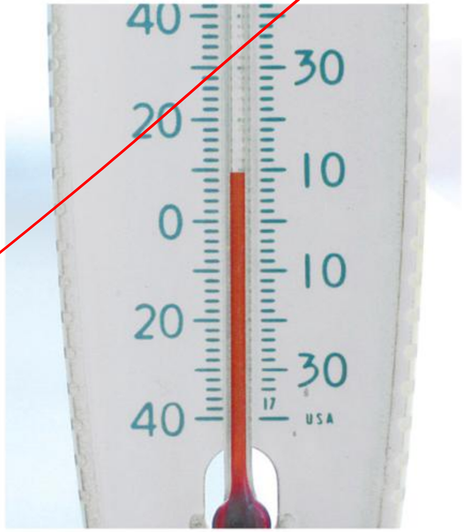
ICE Energy do to "Hot"

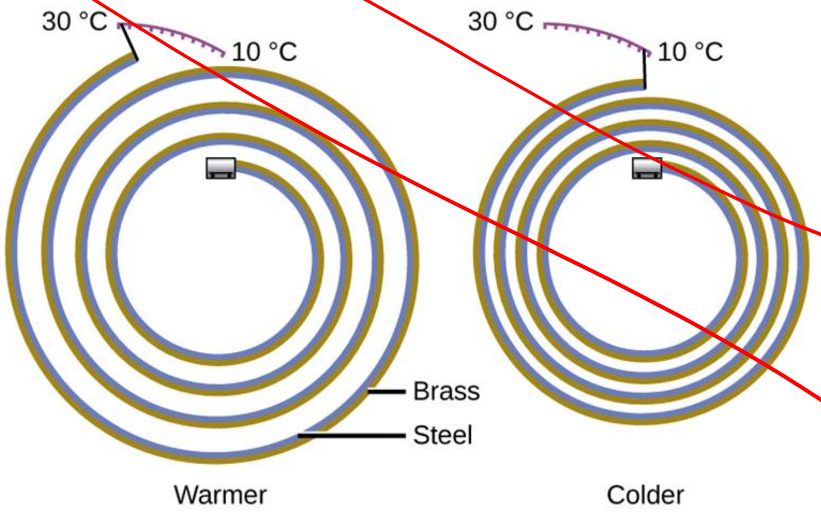
- 30 s

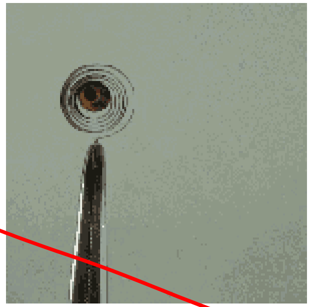
Temperature Up, Most Substances Expand

- This is the property that makes thermometers possible

- Thermometers have a calibrated scale
  - In right thermometer, brass changes more w/ changing temp, causing it to coil/uncoil





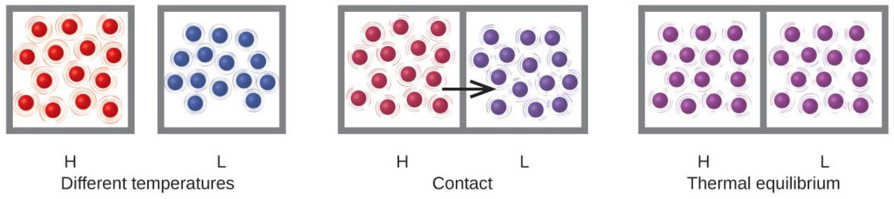


Heat (q)

Energy high temperature low temperature

- **Heat (q)** is the transfer of thermal E from a T_{high} object to a T_{low} object **<u>through contact</u>**.

  - Below, the left **H**igh temp object transfers heat to the **L** temp object
    - The temperature of the high temp object lowers
    - The temperature of the low temp object raises



I **LIKE** this question

- How would you describe the heat transfer when you are cold outside vs when you feel warm outside?

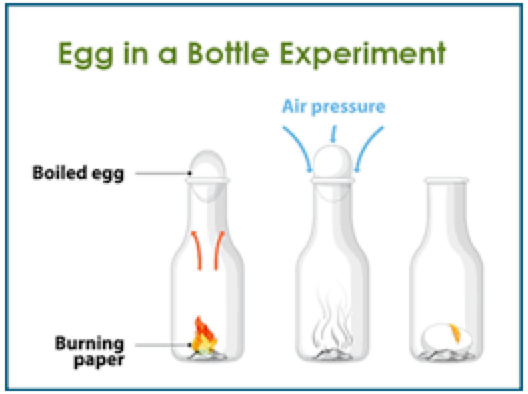
Delete after Spring 2026 semester, moved earlier

Egg on/in a Bottle Experiment (**2 – 3 pt**)

- In Class Demonstration of (ir)relevance of properties changing in Ideal Gas Law write it

$PV = nRT$

- Initially P_{atm} ___=___ $P_{inside\ bottle}$
  - P inside bottle is dependent on g, l, or s?
    - circle above phase.



- Paper lit on fire

1 or 2, Which choice matters?

1. Moles of gas ___↓___
2. Temperature inside bottle ___↑___ → should be P↑

- Observed outcome: $P_{inside\ bottle}$ ___↓___

If we liken the paper to a hydrocarbon (~ $C_{20}H_{42}$)
The reaction is

$C_{20}H_{42}$ (s) + 61 O_2 (g) → 20 CO_2 (g) + 21 H_2O (l)

Equations to Use (Provided on Assessments)

- Equations
- $q = mass * c * \Delta T$ $\Delta T = T_f - T_i$
- $q_{rxn} = -\, mass_{H2O} * 4.18 J / (g * °C) * \Delta T$
- $q_{cal} = C_{cal} * \Delta T$
- $q_{rxn} = -\,(q_{cal} + q_{H2O})$
- $\Delta H° = \Sigma\, \Delta H_f°\,_{products} - \Sigma\, \Delta H_f°\,_{reactants}$
- $\cancel{\Delta E = q + w}$

Principles of Thermochemistry

- In Thermochemistry, we divide the **universe** into a **system** & **surroundings**.

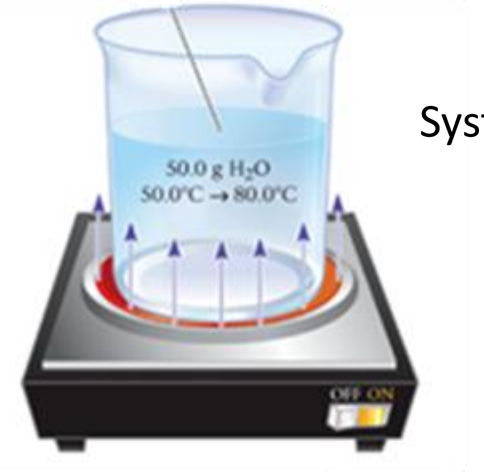
  - **System** – part of the universe that our attention is on.

  - **Surroundings** – rest of the universe. *May* exchange mass or energy with the system.

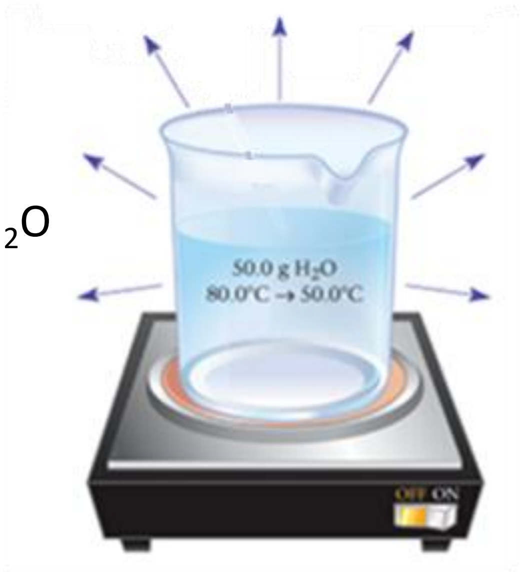
    - In principle, the **surroundings** are mainly the matter in physical contact with the **system**.

Principles of Heat Flow (q)

- **System** – H_2O in beaker
- **Surroundings** – rest of the universe
    - Important part of the surroundings
      =
      the beaker, hotplate, and air above



System = H_2O



Heat flows into water (system)

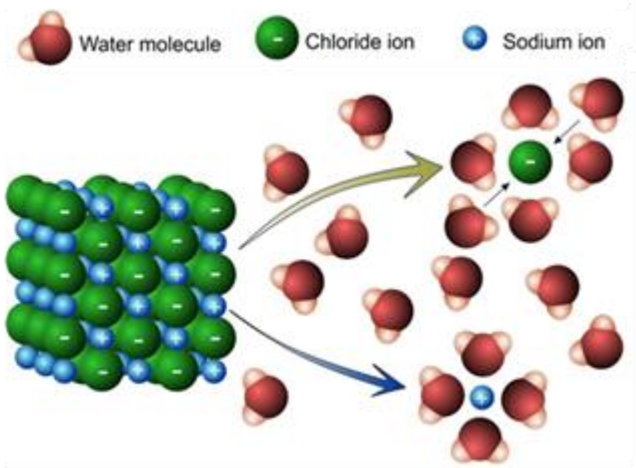
q is positive

Heat flows out of water into air

q is negative

q discussed later

System in a Chemical reaction



- The system in a chemical reaction is composed of
  - Reactants (SM) & Products (Pdt)

$NaCl\ (s) \rightarrow Na^{+}\ (aq) + Cl^{-}\ (aq)$

System includes solid salt and ions

Does **not include** the water that the ions are dissolved in.

- The Surroundings include the reaction vessel (i.e. a beaker), solvent (if present), and air or other material in thermal contact with the system

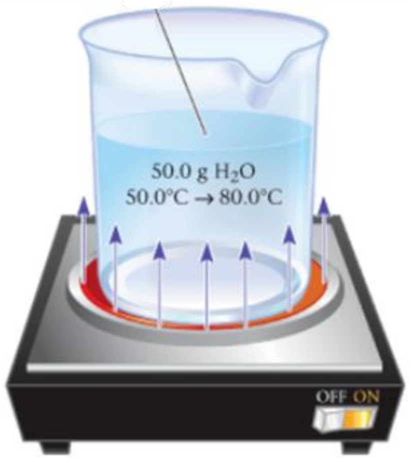
Direction and Sign of **Heat Flow** (**q**)

- q > 0 (is **positive**) when heat flows **into** the **system** from the surroundings

- *Generally,* When heat flows into system, the system's temperature ↑

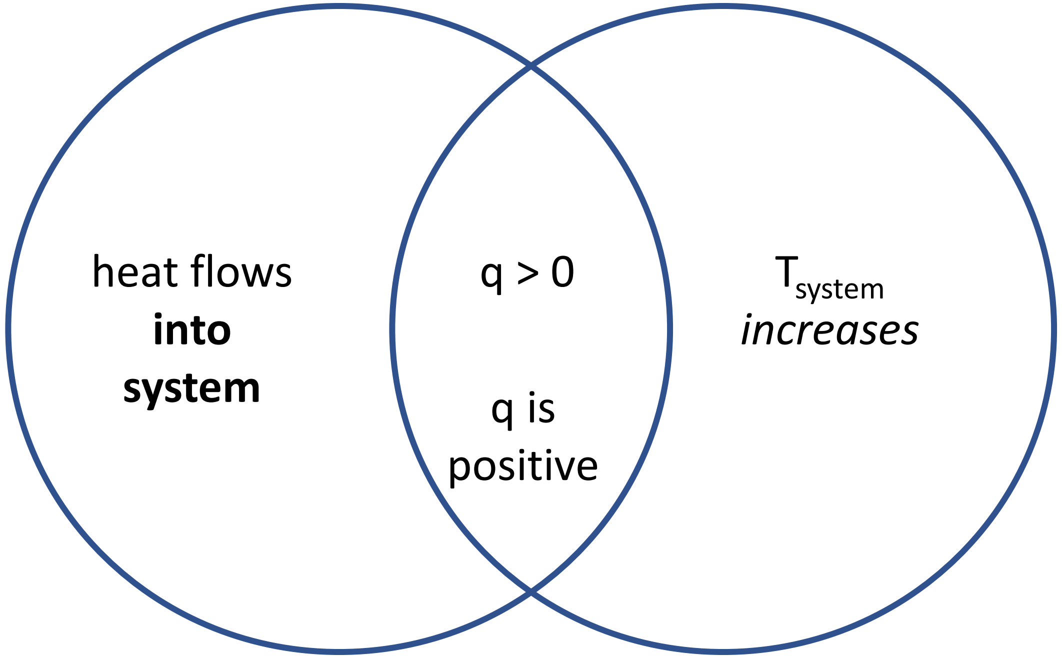
Lowers or rises

- q is like money,
  - When money flows into your system (bank account/wallet), you consider it a positive quantity



Heat flows into water

q is positive

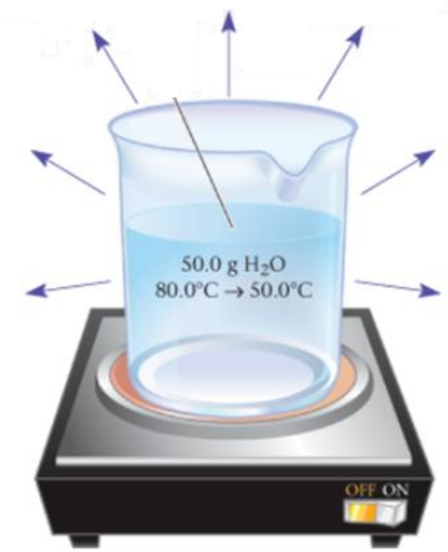

heat flows **into system**
q > 0
q is positive
T_{system} *increases*

Direction and Sign of **Heat Flow** (**q**)

- q < 0 (is **negative**) when heat flows **out of** the system into the surroundings

- *Generally,* When heat flows out of system, the system's temperature lowers

- q is like money,
  - When money flows out of your bank account/wallet (system), you consider it a negative quantity



Heat flows to air from water

q is negative

ICE heat flow of heating water

- 30 s

Endothermic Process

- An **Endothermic** chemical reaction has q > 0 where heat flows from the surroundings into the chemical reaction (system)

- Ice (system) melting: $H_2O\ (s) \rightarrow H_2O\ (l)$ q > 0

Decreases or increases

  - Temperature of surroundings ___↓___

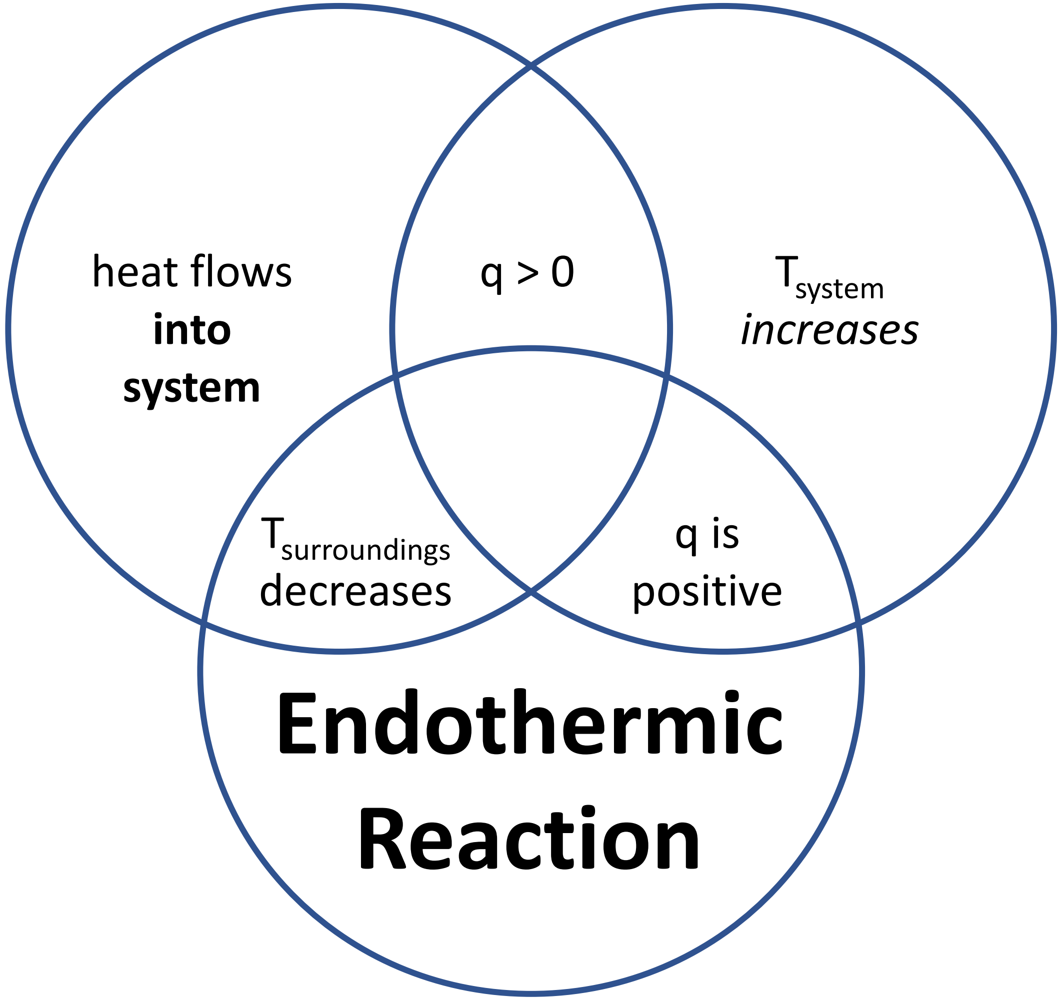
  - breaking chemical bonds ____________ energy. Fill in the blank with releases/absorbs/requires.

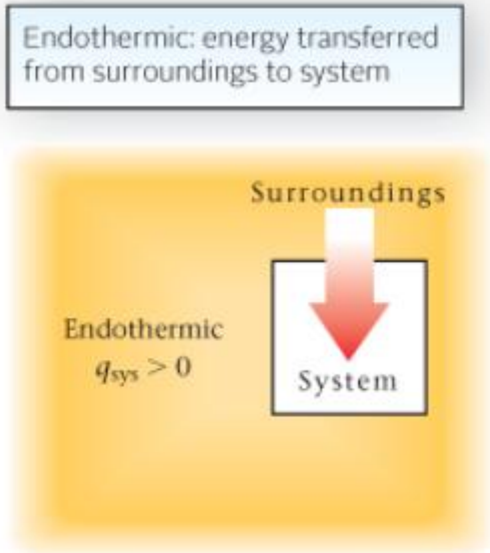
ICE Endothermic

- 30 s

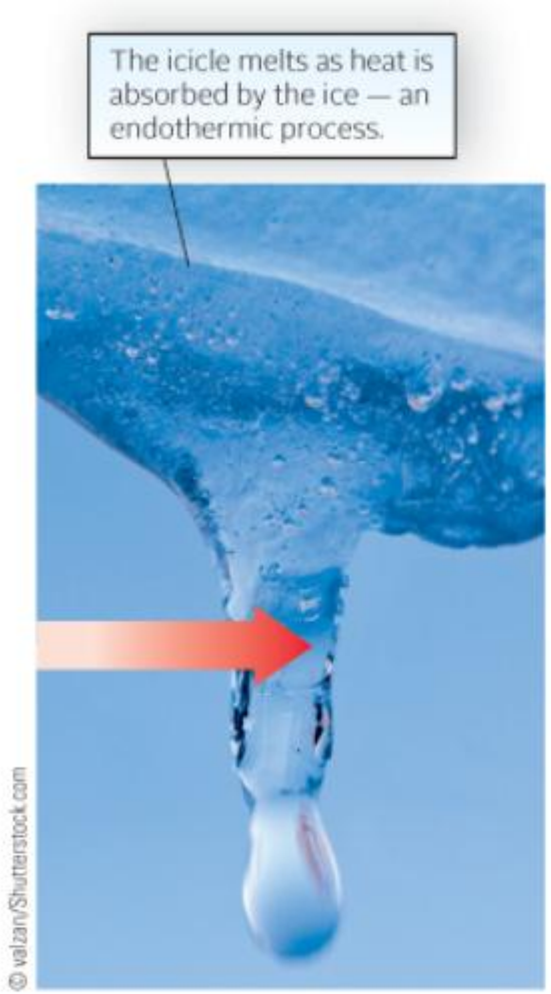
Which of the following phase changes is endothermic?

A) Sublimation	(s) → (g)
B) Deposition	(aq) → (s)
C) Freezing	(l) → (s)
D) Condensation	(g) → (l)





$H_2O\ (s) \rightarrow H_2O\ (l)$

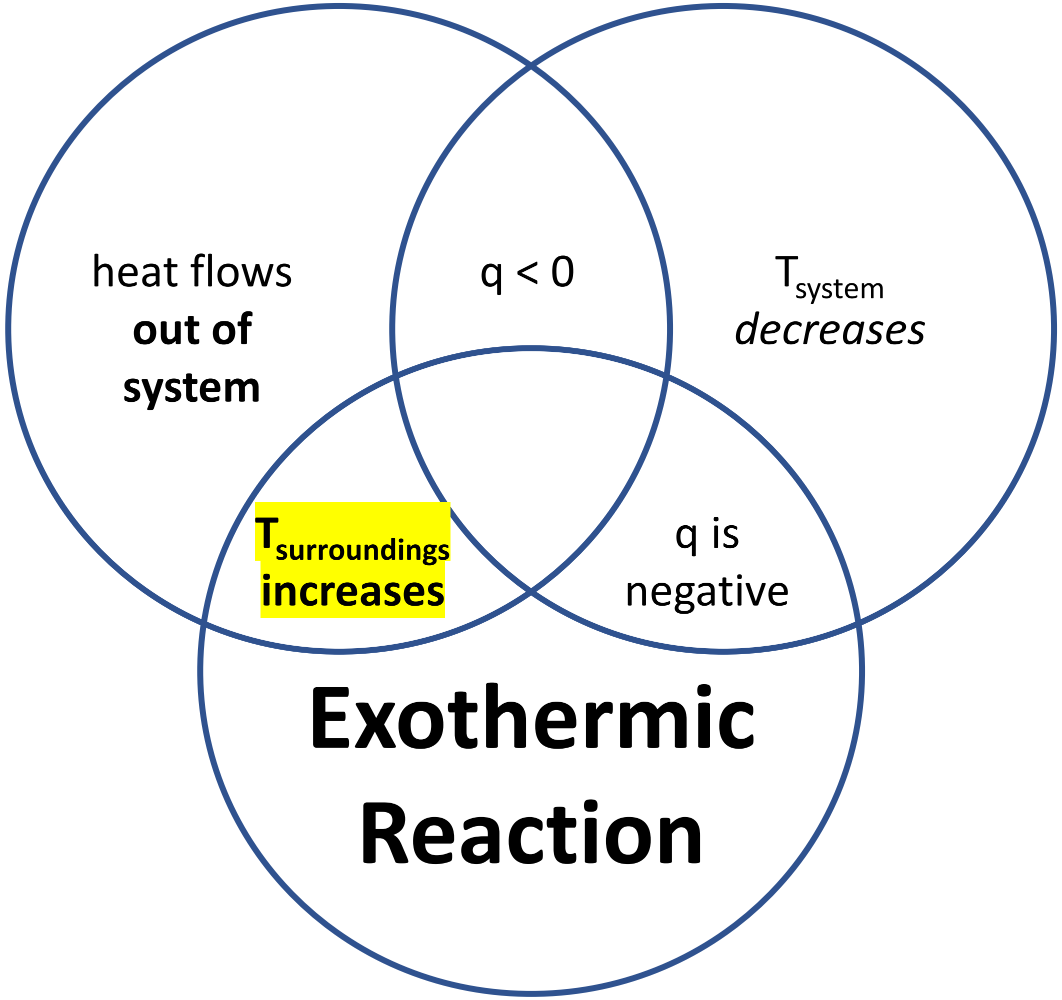


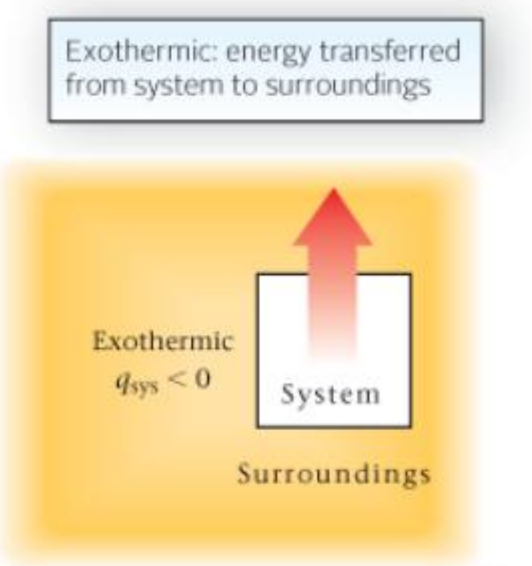
Exothermic Process (think, Explosion)

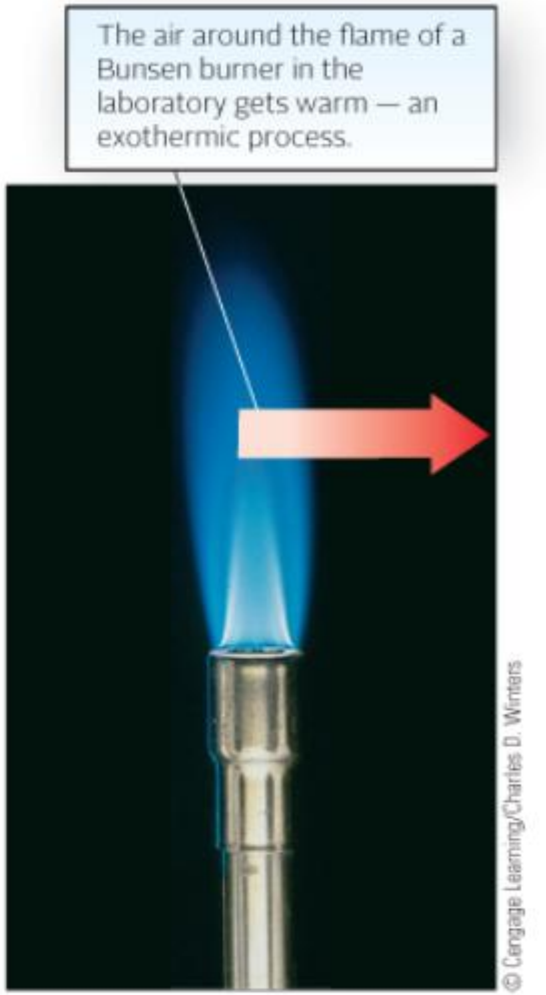
- An **Exothermic** chemical reaction or process has negative q ($q < 0$) where heat flows from the chemical reaction into the surroundings.

- Methane combusting: $CH_4\ (g) + 2\ O_2\ (g) \rightarrow 2H_2O\ (l) + CO_2\ (g)\ q < 0$

  - Temperature of surroundings rises
  - Exothermic reactions are used to (in)directly heat our food.


heat flows
out of
system
q < 0
T system
decreases
T surroundings
increases
q is
negative
Exothermic
Reaction


Exothermic: energy transferred from system to surroundings
Exothermic
q sys < 0
System
Surroundings


The air around the flame of a Bunsen burner in the laboratory gets warm — an exothermic process.
© Cengage Learning/Charles D. Winters

Magnitude of heat flow

- Previously, we considered only the *direction* of **heat flow (q)**

- Energy and q are commonly given in units of joules (J) or kilojoules (kJ)

  - Historically, q was also given in units of calorie (cal)
    - 1 cal is amount of heat required to raise temperature of 1 g H_2O (l) 1°C.
    - 1 cal = 4.184 J, 1 kcal = 4.184 kJ

Calorie in everyday life

- Calorie (Cal) on food labels, is actually a kilocalorie (kcal)
  - 1 Cal = 1 kcal = 1000 cal

- 2,500 – 2,000 Cal/day: recommended daily caloric intake
  - equals 2.5 – 2.0 * 10^6 cal/day.

heat flow (q) and temperature change (ΔT)

- The temperature change (ΔT or ‘delta T’)
- If the only outcome of heat flow is changing the temperature,

$°C \quad °C - °C$

- $q = C * \Delta T$ $\qquad\qquad$ $\Delta T = T_{final} - T_{initial}$

$J = \frac{J}{°C} \; °C$

- C is the **heat capacity** constant of the system and has units of J/ °C (energy per degree Celsius)
    - C is amount of heat required to raise the temperature of the system 1 °C

q & T for a pure substance of fixed mass

- q = C * ΔT
- q = mass * *c* * ΔT

$\Delta T = T_{final} - T_{initial}$

  - **specific heat** or **specific heat capacity** *(c):* J/(g * °C)
    - amount of heat required to raise the temperature of one gram of the substance 1 °C
  - **specific heat** $\frac{J}{g * °C} * \frac{MM\ g\ substance}{1\ mol\ substance} = \frac{J}{mol * °C}$ **molar heat capacity**

- Specific heat is an intensive property (like density, M.P., MF or MM)
  - Recall: mass or volume are extensive properties

Common Specific Heats in J/ (g * °C)

J/ (g * °C)

	c
$Br_2(l)$	0.474
$Cl_2(g)$	0.478
$C_2H_5OH(l)$	2.43
$C_6H_6(l)$	1.72
$CO_2(g)$	0.843
$Cu(s)$	0.382
$Fe(s)$	0.446
$H_2O(g)$	1.87
$H_2O(l)$	4.18
$NaCl(s)$	0.866

Al (s), 0.89 J/ (g * °C)

$$\Delta T = \frac{10\,J}{1\,g\,H_2O \cdot \frac{4.18\,J}{g\,H_2O \cdot °C}} = 2.39°C$$

q = mass * c * ΔT

What does equivalent mean?

For equivalent weights of each of the 4 species labeled in the image, which species temperature raises the most for a given amount of heat being added to the system?

Assume _1_ g of each & 10 J heat transferred (q)

For Cu:

ΔT = q / (mass * c) = 10 / (1 * 0.382) = 26 °C

You should try the other ones!

For these fixed values of mass & q, as c __↑__,
ΔT __↓__.

Practice:

Requires MM of H_2O: 18

$\Delta T = T_{fin} - T_{in}$

$q = mass * c * \Delta T$

J/ (g * °C)

$Br_2(l)$	0.474
$Cl_2(g)$	0.478
$C_2H_5OH(l)$	2.43
$C_6H_6(l)$	1.72
$CO_2(g)$	0.843
$Cu(s)$	0.382
$Fe(s)$	0.446
$H_2O(g)$	1.87
$H_2O(l)$	4.18
$NaCl(s)$	0.866

- Compare the amount of heat, q, given off by
- 1.4 mol of liquid water when it cools from 100 to 30 °C

$q = (1.4_{\cancel{mol}} * \underline{18\ g/\cancel{mol}})\ g\ (4.18\ J/ (\cancel{g} * \cancel{°C})) * (30 - 100)\ \cancel{°C}$

$= \underline{-7374\ J}$ → 7374 J heat put into surroundings

- 1.4 mol of steam (H_2O (g)) cools from 200 to 110 °C.

in fin

$q = (1.4 \cdot 18\ g) \cdot 1.87 \frac{J}{g \cdot °C} \cdot (110 - 200)°C = -4241\ J$

- Which |q| is larger? → for liquid

ICE heat flow

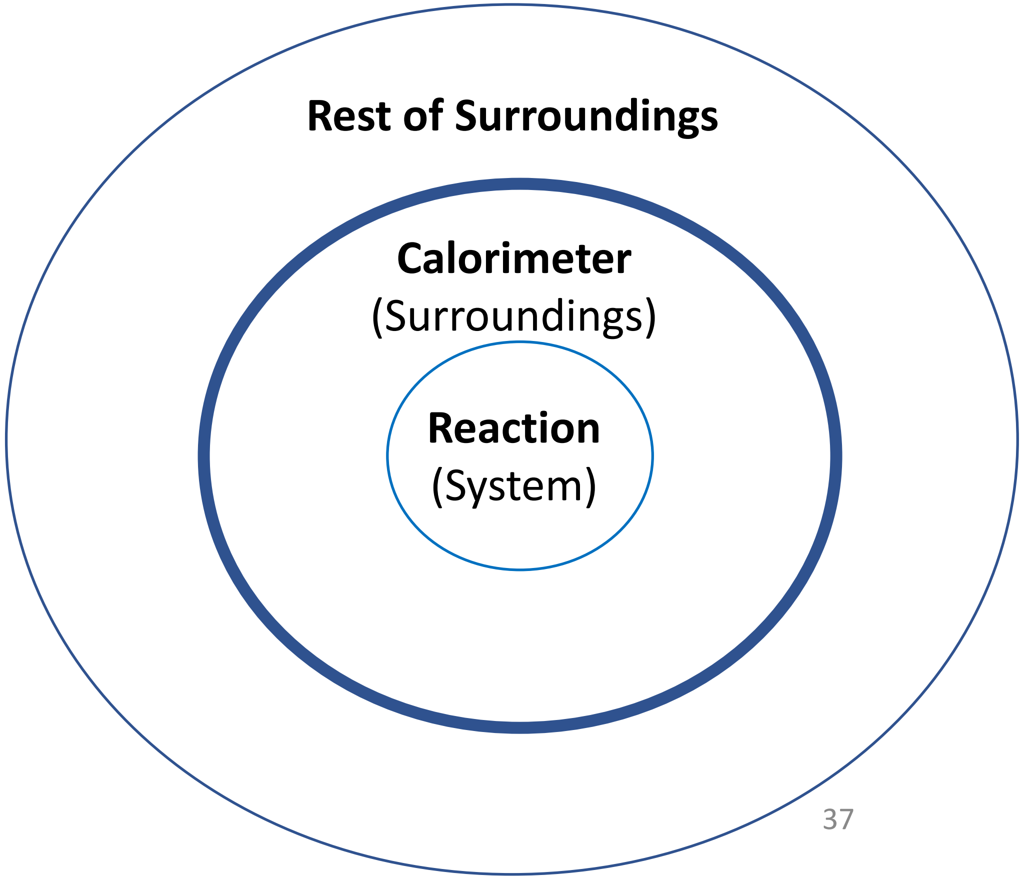
$\Delta T = T_{fin} - T_{in}$

$q = \text{mass} * c * \Delta T$

- 2 mins

Calorimetry – Measurement of Heat Flow

- Calorimeter – device to measure heat flow of a reaction
  - Insulated, such that heat is "*only*" exchanged between calorimeter and reaction (**system**)
  - Thus, heat flow is ***only*** between calorimeter and reaction
  - $\mathbf{q_{rxn} = -\,q_{calorimeter}}$
  - For exothermic reaction ($q_{rxn} < 0$),
    - $q_{calorimeter}$ must be > 0 (positive)
  - $q_{calorimeter} = C_{cal} * \Delta T$



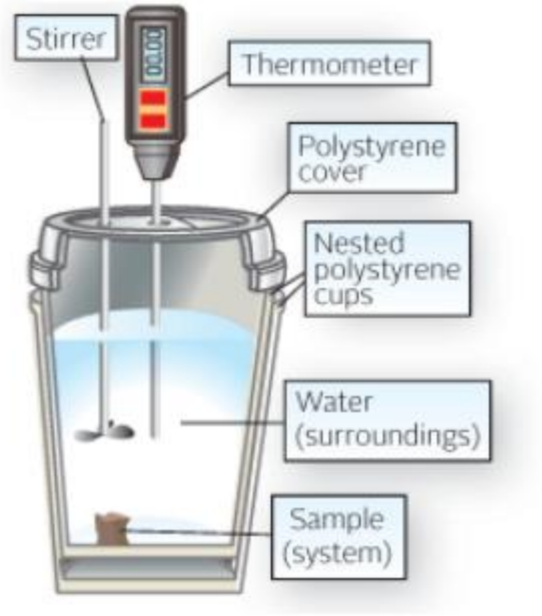
Coffee-Cup Calorimeter (CCC)

- 2 Polystyrene (Styrofoam) cups, good insulators.
  - heat transferred "solely" between H_2O & reaction.

- heat capacity of the calorimeter ~ heat capacity of H_2O inside

- $C_{cal} = mass_{H2O} * c_{H2O} = mass_{water} * 4.18 \frac{J}{g * °C}$
- $q_{reaction} = -q_{calorimeter} = -mass_{H2O} \, 4.18 \frac{J}{g * °C} * \Delta T$

Coffee-Cup Calorimeter.



$q_{reaction} = -\,mass_{H2O}\, 4.18 \frac{J}{g * °C} * \Delta T$

$\Delta T = T_{fin} - T_{in}$

Dissolution reaction – measured by CCC

$CaCl_2$ (s) dissolving is an **endothermic** or **exothermic reaction?** circle correct

- calculate $q_{reaction}$ for 1.0 g of $CaCl_2$ dissolved in 50 g H_2O in a CCC. The temperature of the water rises from 25 to 28.5 °C.

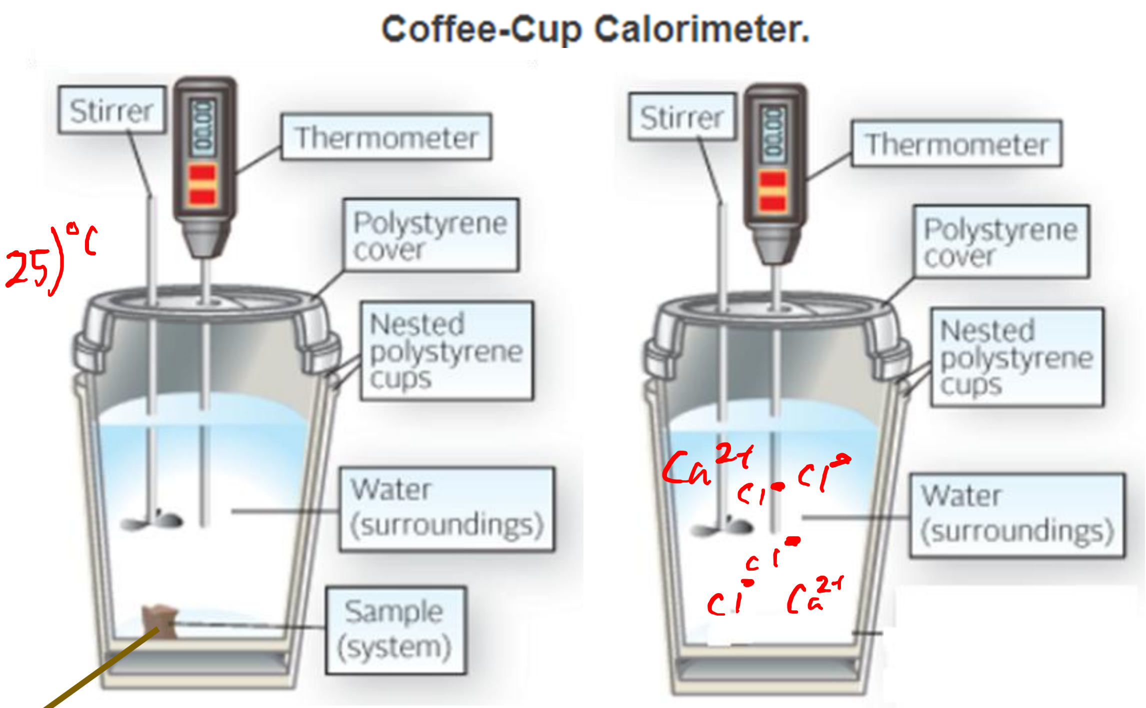
in fin

$q_{rxn} = -50g\left(\frac{4.18J}{g\cdot°C}\right)\cdot(28.5-25)°C$

$= -731.5J$

$\frac{731.5J}{1g\ CaCl_2}$

$CaCl_2\ (s) \rightarrow Ca^{2+}(aq) + 2\ Cl^-(aq)$



- 1.0 g $CaCl_2$ dissolving releases 731.5J

$CaCl_2$ (s)
$CaCl_2$ (s)

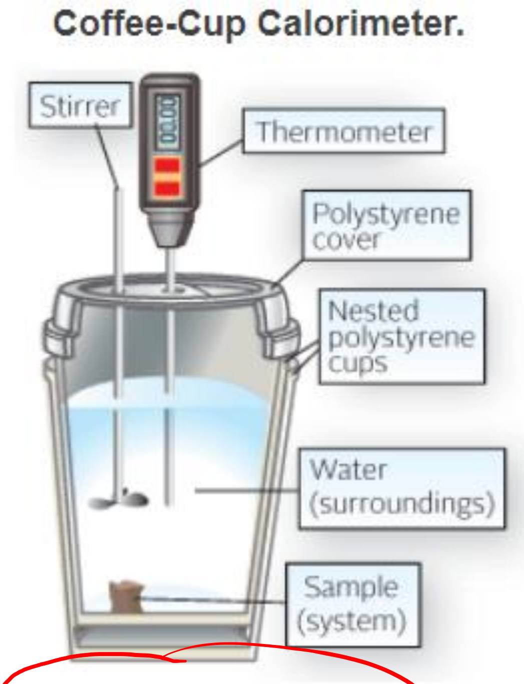
If there were 2 $CaCl_2$ (s) how many ions of each in final? Answer in notes & Drawn in

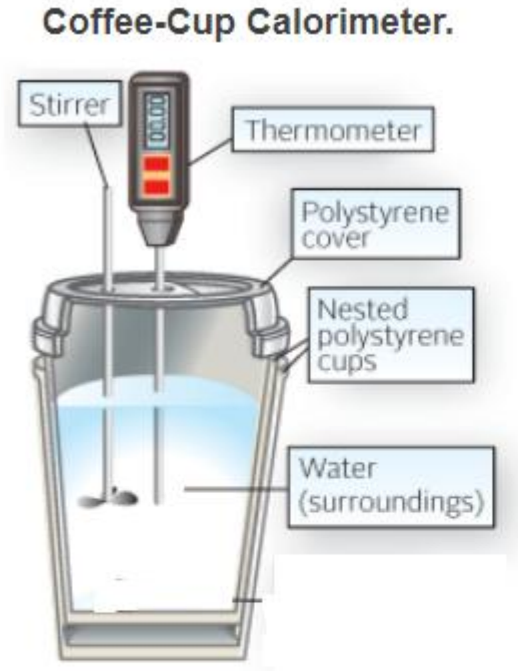
$q_{reaction} = -\,mass_{H2O}\ 4.18\ \frac{J}{g*°C} * \Delta T$

Dissolution reaction – measured by CCC

- How much $CaCl_2$ must be added to 50 g pure water to raise its temperature by 9.00 °C? ΔT
  - Requires calculating q_{rxn} & using the conversion factor from the bottom of last slide!

$CaCl_2\ (s) \rightarrow Ca^{2+}\ (aq) + 2\ Cl^-\ (aq)$





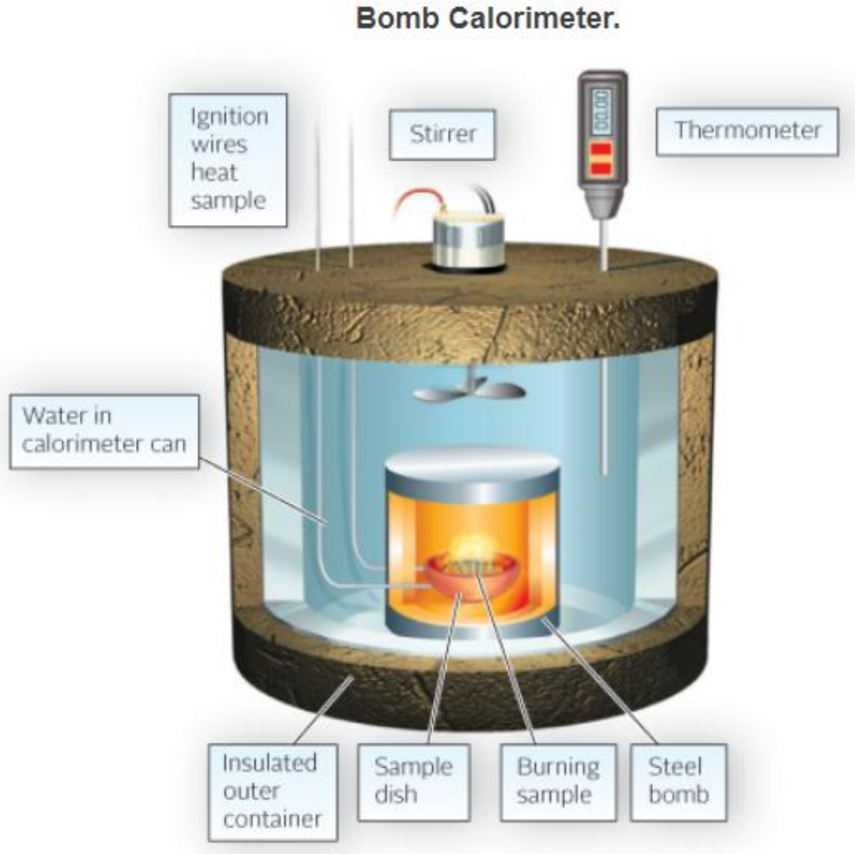
$q_{rxn} = -50g \cdot \frac{4.18J}{g \cdot °C} \cdot 9\ °C = -1881\ J$

$\frac{731.5J}{1g\ CaCl_2}$

$1881J \cdot \frac{1g\ CaCl_2}{731.5J} = 2.57g\ CaCl_2$

Bomb Calorimeter

- CCC's can only be used for reactions that occur in solution and at moderate temperatures, b/c occurs in 2 styrofoam cups.
- determining heat flow (q) for other sorts of reactions can require a bomb calorimeter.
- $q_{reaction} = -(q_{calorimeter} + q_{H2O})$
- $q_{H2O} = mass_{H2O} * 4.18 \frac{J}{g * °C} * \Delta T$
- same ΔT for steel bomb & H_2O
- $q_{calorimeter} = C_{cal} * \Delta T$

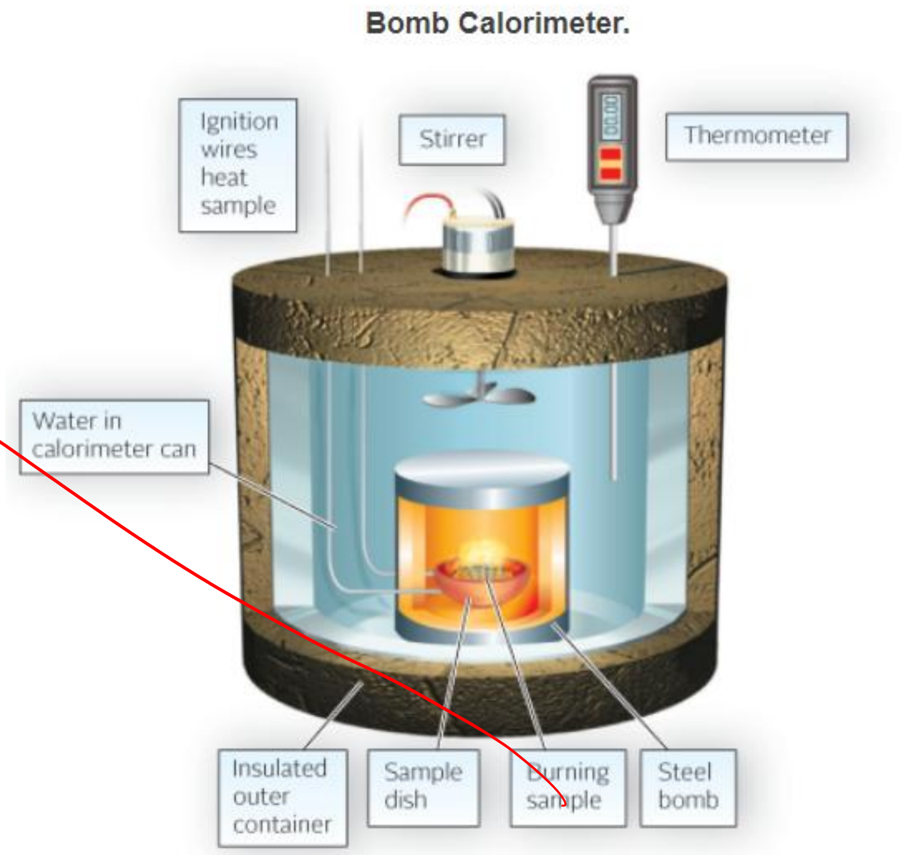


Limiting reactant is *implied to be?*

Example

$$H_2\,(g) + Cl_2\,(g) \rightarrow 2\ HCl\,(g)$$

- When 1 g of H_2 reacts completely with Cl_2 in a bomb calorimeter with a heat capacity of 5.15 kJ/ °C, the temperature w/in the outer container raises from 20 to 30 °C. The calorimeter holds 1.00 kg H_2O. How much heat is evolved in the reaction?
- $q_{reaction} = -(q_{calorimeter} + q_{H2O})$
- $q_{H2O} = mass_{H2O} * 4.18 \frac{J}{g * °C} * \Delta T$
- same <u>ΔT for steel bomb & H_2O</u>
- $q_{calorimeter} = C_{cal} * \Delta T$



The previous slide ends the HW titled
A16: Chapter 9 First Portion.
The subsequent slides for this chapter correspond
to HW titled
A17: Chapter 9 Second Portion

Enthalpy

- **At constant pressure,**
- $q_{reaction} = \Delta H = H_{products} – H_{reactants} = H_{final} – H_{initial}$

- Enthalpy (**H**) – “heat content” or type of chemical energy.
  - State property

    - At constant pressure includes reactions that occur in an open container/beaker/flask as the reaction is exposed to ambient atmosphere which is constant pressure.

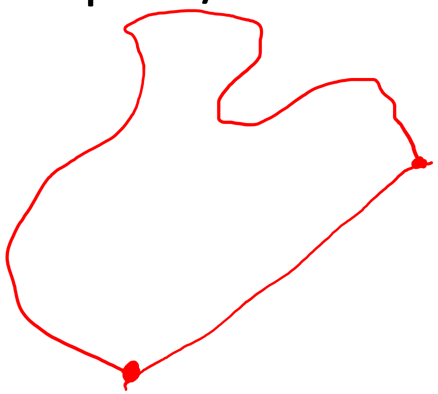
For a chemical reaction:

SM $\rightarrow$ Product

Initial Final

State Properties

- **State Properties** (X) – depend only on the state of the system
  - $\Delta X = X_{final} - X_{initial}$
    - X includes elemental composition, temperature, pressure, volume, mass, enthalpy (H)
  - ***Path-independent***, state properties do not depend on the path/route taken.



- Not state properties:
  - Distance traveled between two cities (*path-dependent*)
  - **Heat flow (q) is not a state property**

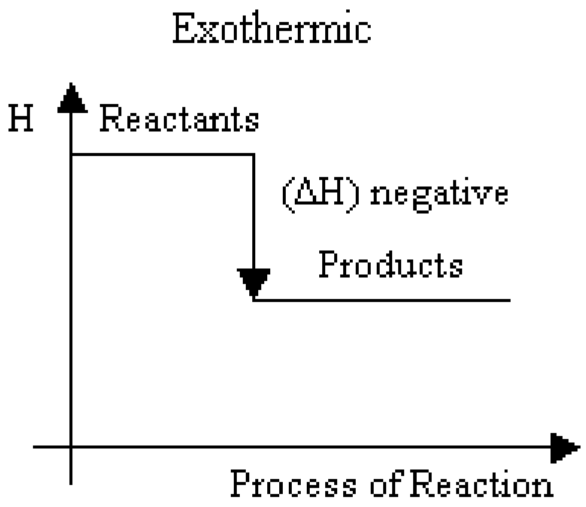
ΔH is negative → heat (q) is produced

Exothermic Reaction ($\Delta H < 0$)

→ Explosion

$q < 0$

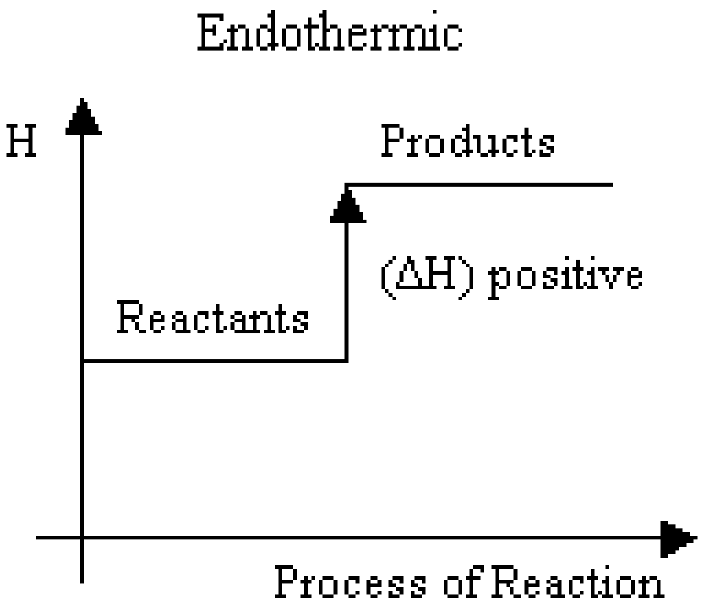
- combustion of methane
  - $CH_4\ (g) + 2\ O_2\ (g) \rightarrow CO_2\ (g) + 2\ H_2O\ (g)$ + heat $\Delta H < 0$
  - **heat is released** from the reaction and the products have lower enthalpy than the reactants



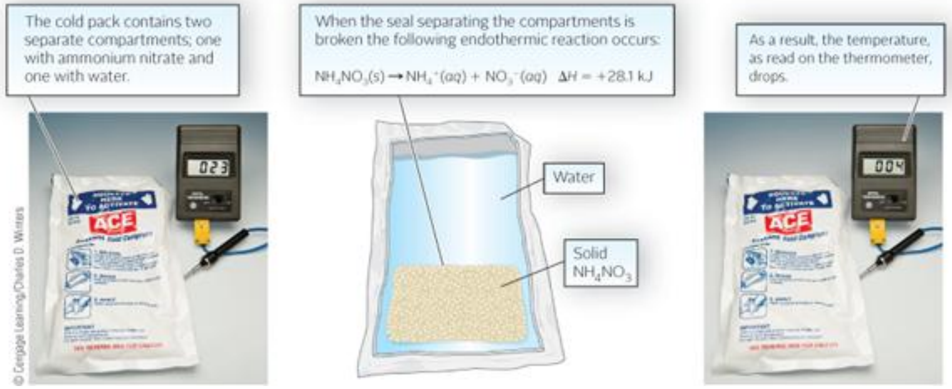
ΔH is (+) & heat is a reactant.

Endothermic Reaction ($\Delta H > 0$)

- Ice melting
  - heat + H_2O (s) → H_2O (l) $\Delta H > 0$
  - q
  - heat is required for the reaction and the products have higher enthalpy than the reactants



Thermochemical Equations



- Thermochemical equation (reaction) – chemical equation that shows enthalpy difference between products and reactants.

q + NH_4NO_3 (s) → NH_4^+ (aq) + NO_3^- (aq) ΔH = + 28.1 kJ

- 28.1 kJ heat entered system to dissolve 1 mol of NH_4NO_3 (s)
- Endothermic

If heat entered system, surroundings __ heat.

Fill in blanks w/ Exothermic or Endothermic

H_2 (g) + Cl_2 (g) → 2 HCl (g) + q (heat) ΔH = – 185 kJ

- 185 kJ heat left system when 2 moles HCl formed from 1 mol of H_2 & 1 mol Cl_2
- Exothermic

- **Must balance chemical equation, and specify phase of matter**

α = 'directly proportional'

Thermochemical Rules

Balanced chemical reactions are in terms of mol

- The magnitude of ΔH is **α** to the molar amount of reactant/product

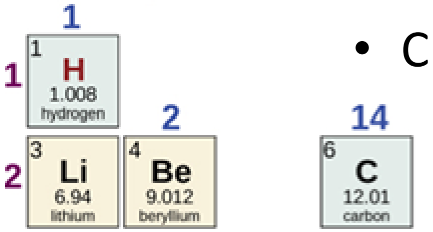
Rxn 1 H_2 (g) + Cl_2 (g) → 2 HCl (g) ΔH_1 = –185 kJ

185 kJ released/mol Cl_2 consumed or for every 2 mol HCl/185 kJ released

Rxn 2 2 H_2 (g) + 2 Cl_2 (g) → 4 HCl (g) ΔH_2 = 2 * (-185 kJ) = -370 kJ

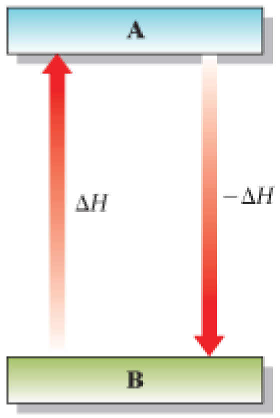
Rxn 3 ½ H_2 (g) + ½ Cl_2 (g) → HCl (g) ΔH_3 = 1/2 * (-185 kJ) = -92.5 kJ

- For the below reaction calculate ΔH when 5.00 g of CH_4 react with an excess of O_2.
  - CH_4 (g) + 2 O_2 (g) → CO_2 (g) + 2 H_2O (l) ΔH = –890.3 kJ



Thermochemical Rules

- ΔH for a reaction is equal in magnitude but opposite in sign for the reverse reaction
- ΔH is a state function



Exothermic

CH_4 (g) + 2 O_2 (g) → CO_2 (g) + 2 H_2O (l) + q ΔH = –890.3 kJ

CO_2 (g) + 2 H_2O (l) → CH_4 (g) + 2 O_2 (g) ΔH = +890.3 kJ

Endothermic

$$q + H_2O(s) \rightarrow H_2O(l) \qquad \Delta H = +6.00\ \text{kJ}$$

$$H_2O(l) \rightarrow H_2O(s) + q \qquad \Delta H = -6.00\ \text{kJ}$$

Practice

Given $2H_2(g) + O_2(g) \rightarrow 2H_2O(l) \quad \Delta H = -571.6\ \mathrm{kJ}$

calculate ΔH for the equation $H_2O(l) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$

Apr 28th

ΔH_{fus} & ΔH_{vap}

- The heat __________ when a solid melts (s → l) is referred to as the heat of fusion (ΔH_{fus})

Released or absorbed?

- The heat __________ when a liquid vaporizes (l → g) is referred to as the heat of vaporization (ΔH_{vap})

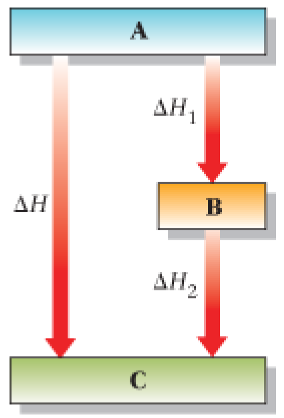
ΔH (kJ/mol) for phase changes

Substance		mp (°C)	ΔH_{fus}*	bp (°C)	ΔH_{vap}*
Benzene	C_6H_6	5	9.84	80	30.8
Bromine	Br_2	−7	10.8	59	29.6
Mercury	Hg	−39	2.33	357	59.4
Naphthalene	$C_{10}H_8$	80	19.3	218	43.3
Water	H_2O	0	6.00	100	40.7

*Values of ΔH_{fus} are given at the melting point, values of ΔH_{vap} at the boiling point. The heat of vaporization of water decreases from 44.9 kJ/mol at 0°C to 44.0 kJ/mol at 25°C to 40.7 kJ/mol at 100°C.

Thermochemical Rules – Hess's law

- The value of ΔH for a reaction is the same whether it occurs in one step or in a series of steps
  - If a thermochemical equation can be expressed as the sum of two or more chemical equations,
    - equation = equation (1) + equation (2) + …
    - $\Delta H = \Delta H_1 + \Delta H_2 + \ldots$
  - Based on Enthalpy (H) being a state property or <u>path-independent</u>.



Practice

Carbon monoxide, CO, is a poisonous gas. It can be obtained by burning carbon in a limited amount of oxygen. Given

(1) $\mathrm{C}(s) + \mathrm{O_2}(g) \rightarrow \mathrm{CO_2}(g) \qquad \Delta H = -393.5\ \mathrm{kJ}$

(2) $2\mathrm{CO}(g) + \mathrm{O_2}(g) \rightarrow 2\mathrm{CO_2}(g) \qquad \Delta H = -566.0\ \mathrm{kJ}$

calculate ΔH for the reaction $\mathrm{C}(s) + \frac{1}{2}\mathrm{O_2}(g) \rightarrow \mathrm{CO}(g) \qquad \Delta H = ?$

Summarizing the rules of thermochemistry:

1. ΔH is directly proportional to the amount of reactant or product.

2. ΔH changes sign when a reaction is reversed.

3. ΔH for a reaction has the same value regardless of the number of steps.

ICE Hess's Law

- 3 mins
  - equation = equation (1) + equation (2) + …
  - $\Delta H = \Delta H_1 + \Delta H_2 + \ldots$

Summarizing the rules of thermochemistry:

1. ΔH is directly proportional to the amount of reactant or product.
2. ΔH changes sign when a reaction is reversed.
3. ΔH for a reaction has the same value regardless of the number of steps.

$$\mathrm{H_2O}(s) \rightarrow \mathrm{H_2O}(l) \qquad \Delta H = +6.00 \text{ kJ}$$

$$\mathrm{H_2O}(l) \rightarrow \mathrm{H_2O}(s) \qquad \Delta H = -6.00 \text{ kJ}$$

H_2 (g) + Cl_2 (g) → 2 HCl (g) $\quad \Delta H_1 = -185$ kJ

2 H_2 (g) + 2 Cl_2 (g) → 4 HCl (g) $\quad \Delta H_2 = 2 * (-185 \text{ kJ}) = -370$ kJ

Standard Molar **Enthalpy of Formation** ($\Delta H_f°$)

- Enthalpy change when 1 mole of the compound is formed at constant pressure of 1 atm and a fixed temperature (generally 25 °C) from the elements in their stable states at that same P&T

$$Ag(s, 25\,°C) + \frac{1}{2}Cl_2(g, 25\,°C, 1\text{ atm}) \rightarrow AgCl(s, 25\,°C) \qquad \Delta H = -127.1\text{ kJ}$$

$$\frac{1}{2}N_2(g, 1\text{ atm}, 25\,°C) + O_2(g, 1\text{ atm}, 25\,°C) \rightarrow NO_2(g, 1\text{ atm}, 25\,°C) \qquad \Delta H = +33.2\text{ kJ}$$

$$\Delta H_f°\ AgCl(s) = -127.1\text{ kJ/mol} \qquad \Delta H_f°\ NO_2(g) = +33.2\text{ kJ/mol}$$

- Requires that elements in stable state at this P&T have $\Delta H_f°$ = 0
  - $\Delta H_f°$ for H^+ is arbitrarily set as 0.

$$\Delta H_f°\ Br_2(l) = \Delta H_f°\ O_2(g) = 0$$

$$\Delta H_f°\ H^+(aq) = 0$$

$\Delta H_f°$ at 25 °C of compounds at 1 atm or (aq) ions at 1 M

Compounds							
$AgBr(s)$	−100.4	$CaCl_2(s)$	−795.8	$H_2O(g)$	−241.8	$NH_4NO_3(s)$	−365.6
$AgCl(s)$	−127.1	$CaCO_3(s)$	−1206.9	$H_2O(l)$	−285.8	$NO(g)$	+90.2
$AgI(s)$	−61.8	$CaO(s)$	−635.1	$H_2O_2(l)$	−187.8	$NO_2(g)$	+33.2
$AgNO_3(s)$	−124.4	$Ca(OH)_2(s)$	−986.1	$H_2S(g)$	−20.6	$N_2O_4(g)$	+9.2
$Ag_2O(s)$	−31.0	$CaSO_4(s)$	−1434.1	$H_2SO_4(l)$	−814.0	$NaCl(s)$	−411.2
$Al_2O_3(s)$	−1675.7	$CdCl_2(s)$	−391.5	$HgO(s)$	−90.8	$NaF(s)$	−573.6
$BaCl_2(s)$	−858.6	$CdO(s)$	−258.2	$KBr(s)$	−393.8	$NaOH(s)$	−425.6
$BaCO_3(s)$	−1216.3	$Cr_2O_3(s)$	−1139.7	$KCl(s)$	−436.7	$NiO(s)$	−239.7
$BaO(s)$	−553.5	$CuO(s)$	−157.3	$KClO_3(s)$	−397.7	$PbBr_2(s)$	−278.7
$BaSO_4(s)$	−1473.2	$Cu_2O(s)$	−168.6	$KClO_4(s)$	−432.8	$PbCl_2(s)$	−359.4
$CCl_4(l)$	−135.4	$CuS(s)$	−53.1	$KNO_3(s)$	−494.6	$PbO(s)$	−219.0
$CHCl_3(l)$	−134.5	$Cu_2S(s)$	−79.5	$MgCl_2(s)$	−641.3	$PbO_2(s)$	−277.4
$CH_4(g)$	−74.8	$CuSO_4(s)$	−771.4	$MgCO_3(s)$	−1095.8	$PCl_3(g)$	−287.0
$C_2H_2(g)$	+226.7	$Fe(OH)_3(s)$	−823.0	$MgO(s)$	−601.7	$PCl_5(g)$	−374.9
$C_2H_4(g)$	+52.3	$Fe_2O_3(s)$	−824.2	$Mg(OH)_2(s)$	−924.5	$SiO_2(s)$	−910.9
$C_2H_6(g)$	−84.7	$Fe_3O_4(s)$	−1118.4	$MgSO_4(s)$	−1284.9	$SnO_2(s)$	−580.7
$C_3H_8(g)$	−103.8	$HBr(g)$	−36.4	$MnO(s)$	−385.2	$SO_2(g)$	−296.8
$CH_3OH(l)$	−238.7	$HCl(g)$	−92.3	$MnO_2(s)$	−520.0	$SO_3(g)$	−395.7
$C_2H_5OH(l)$	−277.7	$HF(g)$	−271.1	$NH_3(g)$	−46.1	$ZnI_2(s)$	−208.0
$CO(g)$	−110.5	$H\ (g)$	+26.5	$N_2H_4(l)$	+50.6	$ZnO(s)$	−348.3
$CO_2(g)$	−393.5	$HNO_3(l)$	−174.1	$NH_4Cl(s)$	−314.4	$ZnS(s)$	−206.0

Cations				Anions			
$Ag^+(aq)$	+105.6	$Hg^{2+}(aq)$	+171.1	$Br^-(aq)$	−121.6	$HPO_4^{2-}(aq)$	−1292.1
$Al^{3+}(aq)$	−531.0	$K^+(aq)$	−252.4	$CO_3^{2-}(aq)$	−677.1	$HSO_4^-(aq)$	−887.3
$Ba^{2+}(aq)$	−537.6	$Mg^{2+}(aq)$	−466.8	$Cl^-(aq)$	−167.2	$I^-(aq)$	−55.2
$Ca^{2+}(aq)$	−542.8	$Mn^{2+}(aq)$	−220.8	$ClO_3^-(aq)$	−104.0	$MnO_4^-(aq)$	−541.4
$Cd^{2+}(aq)$	−75.9	$Na^+(aq)$	−240.1	$ClO_4^-(aq)$	−129.3	$NO_2^-(aq)$	−104.6
$Cu^+(aq)$	+71.7	$NH_4^+(aq)$	−132.5	$CrO_4^{2-}(aq)$	−881.2	$NO_3^-(aq)$	−205.0
$Cu^{2+}(aq)$	+64.8	$Ni^{2+}(aq)$	−54.0	$Cr_2O_7^{2-}(aq)$	−1490.3	$OH^-(aq)$	−230.0
$Fe^{2+}(aq)$	−89.1	$Pb^{2+}(aq)$	−1.7	$F^-(aq)$	−332.6	$PO_4^{3-}(aq)$	−1277.4
$Fe^{3+}(aq)$	−48.5	$Sn^{2+}(aq)$	−8.8	$HCO_3^-(aq)$	−692.0	$S^{2-}(aq)$	+33.1
$H^+(aq)$	0.0	$Zn^{2+}(aq)$	−153.9	$H_2PO_4^-(aq)$	−1296.3	$SO_4^{2-}(aq)$	−909.3

Calculating $\Delta H°$ from $\Delta H_f°$

- $\Delta H° = \Sigma\ [\Delta H_f°\ \text{products}] - \Sigma\ [\Delta H_f°\ \text{reactants}]$

$$2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(s) + Al_2O_3(s)$$

$$\Delta H° = \Sigma\, \Delta H_f°\ \text{products} - \Sigma\, \Delta H_f°\ \text{reactants} = \Delta H_f°\ Al_2O_3(s) - \Delta H_f°\ Fe_2O_3(s)$$

  - Must account for # of moles of each species (stoichiometric coefficients)

$$2Al(s) + 3Cu^{2+}(aq) \rightarrow 2Al^{3+}(aq) + 3Cu(s)$$

$$\Delta H° = 2\Delta H_f°\ Al^{3+}(aq) - 3\Delta H_f°\ Cu^{2+}(aq)$$

Practice – Requires $\Delta H_f°$ Slide

Benzene, C_6H_6, used in the manufacture of plastics, is a carcinogen affecting the bone marrow. Long-term exposure has been shown to cause leukemia and other blood disorders. The combustion of benzene is given by the following equation:

$$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l) \qquad \Delta H° = -3267.4\ \text{kJ}$$

a Calculate the heat of formation of benzene.

$\Delta H° = \Sigma\ [\Delta H_f°\ \text{products}] - \Sigma\ [\Delta H_f°\ \text{reactants}]$

$-3267.4\ \text{kJ} = [6\ \text{mol} * (-393.5\ \text{kJ/mol}) + 3\ \text{mol}\ (-285.8\ \text{kJ/mol})] – [\Delta H_f°{}_{benzene} + 15/2\ \text{mol}\ (0\ \text{kJ/mol})]$

Solve for $\Delta H_f°{}_{benzene}$

Practice – built of $\Delta H_f°$ Slide & previous slide

Benzene, C_6H_6, used in the manufacture of plastics, is a carcinogen affecting the bone marrow. Long-term exposure has been shown to cause leukemia and other blood disorders. The combustion of benzene is given by the following equation:

$$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l) \qquad \Delta H° = -3267.4\ \text{kJ}$$

b Calculate $\Delta H°$ for the reaction

$$12CO_2(g) + 6H_2O(l) \rightarrow 2C_6H_6(l) + 15O_2(g)$$

Practice

Copper carbonate is used to control algae in farm ponds. When hydrochloric acid is added to a solution of copper carbonate, carbon dioxide gas is formed (**Figure 8.13**). The equation for the reaction is

$$2H^{+}(aq) + CO_3^{2-}(aq) \rightarrow CO_2(g) + H_2O(l)$$

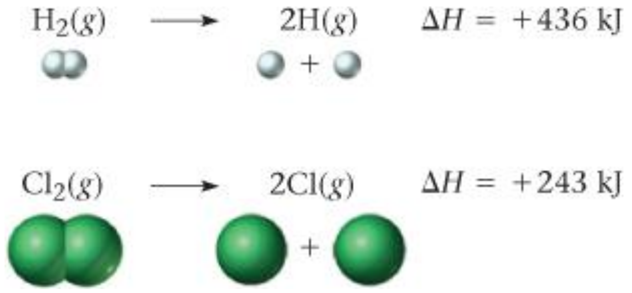
a Calculate ΔH° for the thermochemical equation.

$\Delta H^\circ = \Sigma\,[\Delta H_f^\circ \text{ products}] - \Sigma\,[\Delta H_f^\circ \text{ reactants}]$

$2\ H\ (g) \rightarrow H_2\ (g)$ ΔH = ______

Bond Enthalpy

- Bond Enthalpy ΔH for 1 mole of bonds broken in the gaseous state
  - 1 mole of H – H bonds broken: +436 kJ



- Breaking bonds requires heat (energy) to be added to system
  - Endothermic process

Bond Enthalpies

Single Bond Enthalpy (kJ/mol)									
	H	C	N	O	S	F	Cl	Br	I
H	436	414	389	464	339	565	431	368	297
C		347	293	351	259	485	331	276	218
N			159	222	—	272	201	243	—
O				138	—	184	205	201	201
S					226	285	255	213	—
F						153	255	255	277
Cl							243	218	209
Br								193	180
I									151

Multiple Bond Enthalpy (kJ/mol)					
C=C	612	N=N	418	C≡C	820
C=N	615	N=O	607	C≡N	890
C=O	715	O=O	498	C≡O	1075
C=S	477	S=O	498	N≡N	941

Bond formation – heat is released

$$H(g) + Cl(g) \longrightarrow HCl(g) \quad \Delta H = -431 \text{ kJ}$$

$$H(g) + F(g) \longrightarrow HF(g) \quad \Delta H = -565 \text{ kJ}$$

Bond enthalpies can explain ΔH_{rxn}

- Endothermic reaction if
  - Reactant bonds are stronger than product bonds
  - There are more bonds in the reactants than products

- For increasing number of bonds between same atoms, bond enthalpy increases

$$\text{bond enthalpy C—C} = 347\ \text{kJ/mol}$$

$$\text{bond enthalpy C=C} = 612\ \text{kJ/mol}$$

$$\text{bond enthalpy C}\equiv\text{C} = 820\ \text{kJ/mol}$$

Chopsticks as bonds

The First Law of Thermodynamics

- Thermochemistry is a branch of **thermodynamics**, which deals with processes pertaining to energy
  - In Thermodynamics there are two types of energy, heat (q) or work (w)
    - **Work** (**w**) is all forms of energy other than heat

- The First Law of Thermodynamics – Energy (**E**) cannot be created or destroyed; it can only be transferred (between system & surroundings).
  - $\Delta E_{system} = -\Delta E_{surroundings}$
  - $\Delta E_{system} = \Delta E = q + w$
      - q & w positive when heat and work enter the system from the surroundings (same as early)

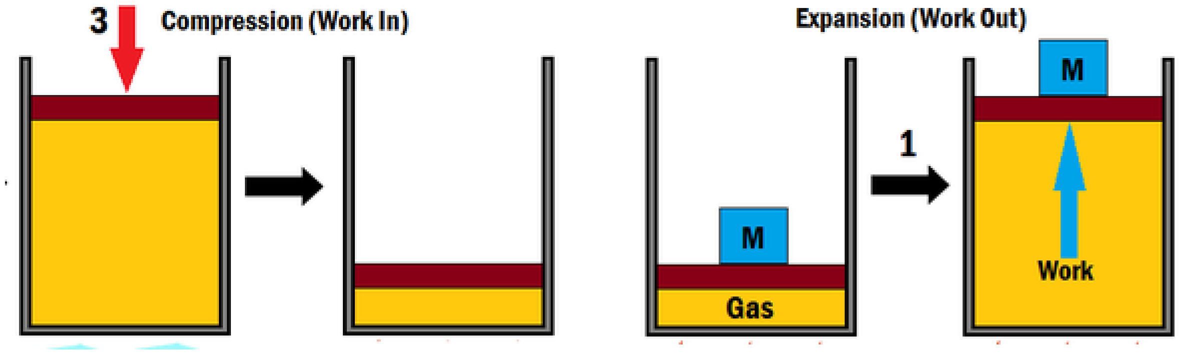
Energy & pressurized gas are “Danger”!!!

- Pressurized gas has a lot of energy in that system
  - energy had to be spent compressing it all into the small confines it is in.
  - If the energy is utilized in uncontrolled pathways, it can be dire!

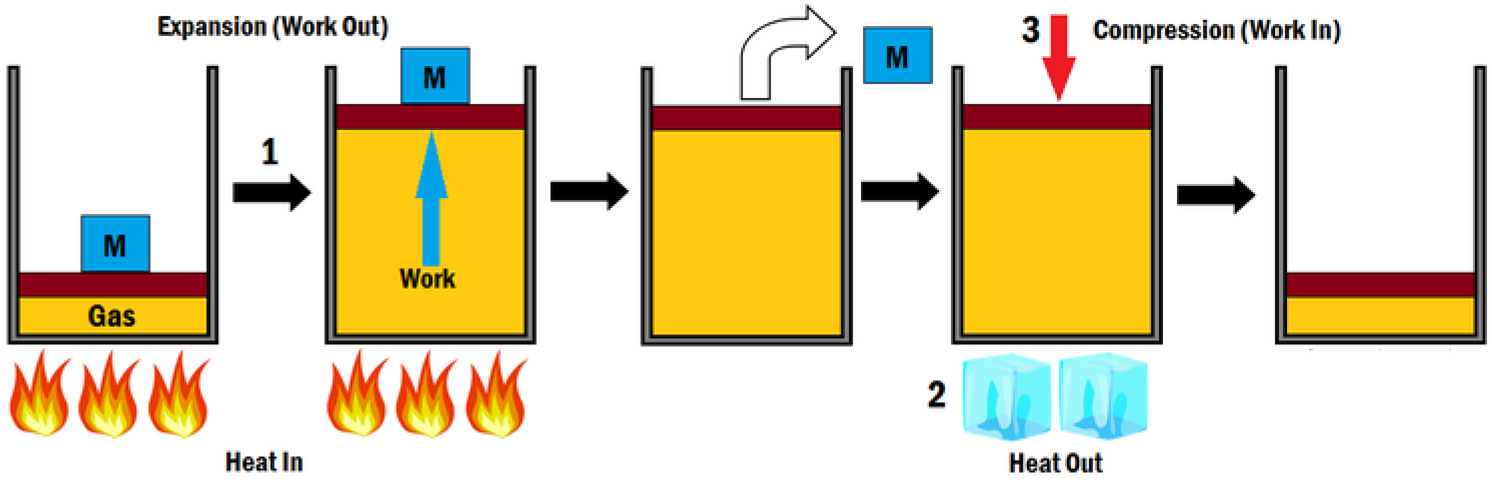
The First Law of Thermodynamics

- **Work** (**w**) is all forms of energy other than heat
- $\Delta E_{system} = \Delta E = q + w$
    - q & w positive when heat and work enter the system from the surroundings (same as early)
        - **Has the system increased in energy?**
        - Pressurized gas is higher energy than unpressurized gas
  - Gas (yellow) is the system

w is positive (work being done on system)
-pressurized



w is negative (work being done on surroundings)



q > 0 (heat into system)
w < 0

q < 0 (heat out of system)
w > 0

Practice: Calculate ΔE for a gas that

- Absorbs 20 J of heat and does 12 J of work by expanding

  $\Delta E = q + w$

- Evolves 30 J of heat and has 52 J of work done on it as it contracts.

  $\Delta E = q + w$

$\Delta H = \Delta E + RT\,\Delta n_g$

- Relates change in enthalpy to change in energy for a system
  - Gas constant R = 8.31 J/ (mol * K)
  - Temperature in Kelvin
  - $\Delta n_g = n_{\text{gaseous products}} - n_{\text{gaseous reactants}}$
    - n = # of moles

Practice:

- $\Delta H = \Delta E + RT\,\Delta n_g$
  - Gas constant R = 8.31 J/ (mol * K)   Temperature in Kelvin
  - $\Delta n_g = n_{\text{gaseous products}} - n_{\text{gaseous reactants}}$
    - n = # of moles
  - Requires ΔH_f° (copied to next slide)

Calculate ΔH and ΔE at $25\,^\circ\mathrm{C}$ for the reaction that takes place when an oxyacetylene torch is used.

$$C_2H_2(g) + \frac{5}{2}O_2(g) \rightarrow 2CO_2(g) + H_2O(g)$$

$\Delta H_f°$ at 25 °C of compounds at 1 atm or (aq) ions at 1 M

Compounds

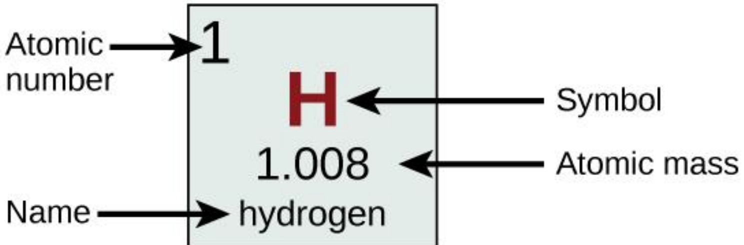
$AgBr(s)$	−100.4	$CaCl_2(s)$	−795.8	$H_2O(g)$	−241.8	$NH_4NO_3(s)$	−365.6
$AgCl(s)$	−127.1	$CaCO_3(s)$	−1206.9	$H_2O(l)$	−285.8	$NO(g)$	+90.2
$AgI(s)$	−61.8	$CaO(s)$	−635.1	$H_2O_2(l)$	−187.8	$NO_2(g)$	+33.2
$AgNO_3(s)$	−124.4	$Ca(OH)_2(s)$	−986.1	$H_2S(g)$	−20.6	$N_2O_4(g)$	+9.2
$Ag_2O(s)$	−31.0	$CaSO_4(s)$	−1434.1	$H_2SO_4(l)$	−814.0	$NaCl(s)$	−411.2
$Al_2O_3(s)$	−1675.7	$CdCl_2(s)$	−391.5	$HgO(s)$	−90.8	$NaF(s)$	−573.6
$BaCl_2(s)$	−858.6	$CdO(s)$	−258.2	$KBr(s)$	−393.8	$NaOH(s)$	−425.6
$BaCO_3(s)$	−1216.3	$Cr_2O_3(s)$	−1139.7	$KCl(s)$	−436.7	$NiO(s)$	−239.7
$BaO(s)$	−553.5	$CuO(s)$	−157.3	$KClO_3(s)$	−397.7	$PbBr_2(s)$	−278.7
$BaSO_4(s)$	−1473.2	$Cu_2O(s)$	−168.6	$KClO_4(s)$	−432.8	$PbCl_2(s)$	−359.4
$CCl_4(l)$	−135.4	$CuS(s)$	−53.1	$KNO_3(s)$	−494.6	$PbO(s)$	−219.0
$CHCl_3(l)$	−134.5	$Cu_2S(s)$	−79.5	$MgCl_2(s)$	−641.3	$PbO_2(s)$	−277.4
$CH_4(g)$	−74.8	$CuSO_4(s)$	−771.4	$MgCO_3(s)$	−1095.8	$PCl_3(g)$	−287.0
$C_2H_2(g)$	+226.7	$Fe(OH)_3(s)$	−823.0	$MgO(s)$	−601.7	$PCl_5(g)$	−374.9
$C_2H_4(g)$	+52.3	$Fe_2O_3(s)$	−824.2	$Mg(OH)_2(s)$	−924.5	$SiO_2(s)$	−910.9
$C_2H_6(g)$	−84.7	$Fe_3O_4(s)$	−1118.4	$MgSO_4(s)$	−1284.9	$SnO_2(s)$	−580.7
$C_3H_8(g)$	−103.8	$HBr(g)$	−36.4	$MnO(s)$	−385.2	$SO_2(g)$	−296.8
$CH_3OH(l)$	−238.7	$HCl(g)$	−92.3	$MnO_2(s)$	−520.0	$SO_3(g)$	−395.7
$C_2H_5OH(l)$	−277.7	$HF(g)$	−271.1	$NH_3(g)$	−46.1	$ZnI_2(s)$	−208.0
$CO(g)$	−110.5	$H\ (g)$	+26.5	$N_2H_4(l)$	+50.6	$ZnO(s)$	−348.3
$CO_2(g)$	−393.5	$HNO_3(l)$	−174.1	$NH_4Cl(s)$	−314.4	$ZnS(s)$	−206.0

Cations				Anions			
$Ag^+(aq)$	+105.6	$Hg^{2+}(aq)$	+171.1	$Br^-(aq)$	−121.6	$HPO_4^{2-}(aq)$	−1292.1
$Al^{3+}(aq)$	−531.0	$K^+(aq)$	−252.4	$CO_3^{2-}(aq)$	−677.1	$HSO_4^-(aq)$	−887.3
$Ba^{2+}(aq)$	−537.6	$Mg^{2+}(aq)$	−466.8	$Cl^-(aq)$	−167.2	$I^-(aq)$	−55.2
$Ca^{2+}(aq)$	−542.8	$Mn^{2+}(aq)$	−220.8	$ClO_3^-(aq)$	−104.0	$MnO_4^-(aq)$	−541.4
$Cd^{2+}(aq)$	−75.9	$Na^+(aq)$	−240.1	$ClO_4^-(aq)$	−129.3	$NO_2^-(aq)$	−104.6
$Cu^+(aq)$	+71.7	$NH_4^+(aq)$	−132.5	$CrO_4^{2-}(aq)$	−881.2	$NO_3^-(aq)$	−205.0
$Cu^{2+}(aq)$	+64.8	$Ni^{2+}(aq)$	−54.0	$Cr_2O_7^{2-}(aq)$	−1490.3	$OH^-(aq)$	−230.0
$Fe^{2+}(aq)$	−89.1	$Pb^{2+}(aq)$	−1.7	$F^-(aq)$	−332.6	$PO_4^{3-}(aq)$	−1277.4
$Fe^{3+}(aq)$	−48.5	$Sn^{2+}(aq)$	−8.8	$HCO_3^-(aq)$	−692.0	$S^{2-}(aq)$	+33.1
$H^+(aq)$	0.0	$Zn^{2+}(aq)$	−153.9	$H_2PO_4^-(aq)$	−1296.3	$SO_4^{2-}(aq)$	−909.3

Periodic Table of the Elements

Period \ Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008 hydrogen																	2 He 4.003 helium
2	3 Li 6.94 lithium	4 Be 9.012 beryllium											5 B 10.81 boron	6 C 12.01 carbon	7 N 14.01 nitrogen	8 O 16.00 oxygen	9 F 19.00 fluorine	10 Ne 20.18 neon
3	11 Na 22.99 sodium	12 Mg 24.31 magnesium											13 Al 26.98 aluminum	14 Si 28.09 silicon	15 P 30.97 phosphorus	16 S 32.06 sulfur	17 Cl 35.45 chlorine	18 Ar 39.95 argon
4	19 K 39.10 potassium	20 Ca 40.08 calcium	21 Sc 44.96 scandium	22 Ti 47.87 titanium	23 V 50.94 vanadium	24 Cr 52.00 chromium	25 Mn 54.94 manganese	26 Fe 55.85 iron	27 Co 58.93 cobalt	28 Ni 58.69 nickel	29 Cu 63.55 copper	30 Zn 65.38 zinc	31 Ga 69.72 gallium	32 Ge 72.63 germanium	33 As 74.92 arsenic	34 Se 78.97 selenium	35 Br 79.90 bromine	36 Kr 83.80 krypton
5	37 Rb 85.47 rubidium	38 Sr 87.62 strontium	39 Y 88.91 yttrium	40 Zr 91.22 zirconium	41 Nb 92.91 niobium	42 Mo 95.95 molybdenum	43 Tc [97] technetium	44 Ru 101.1 ruthenium	45 Rh 102.9 rhodium	46 Pd 106.4 palladium	47 Ag 107.9 silver	48 Cd 112.4 cadmium	49 In 114.8 indium	50 Sn 118.7 tin	51 Sb 121.8 antimony	52 Te 127.6 tellurium	53 I 126.9 iodine	54 Xe 131.3 xenon
6	55 Cs 132.9 cesium	56 Ba 137.3 barium	57-71 La–Lu *	72 Hf 178.5 hafnium	73 Ta 180.9 tantalum	74 W 183.8 tungsten	75 Re 186.2 rhenium	76 Os 190.2 osmium	77 Ir 192.2 iridium	78 Pt 195.1 platinum	79 Au 197.0 gold	80 Hg 200.6 mercury	81 Tl 204.4 thallium	82 Pb 207.2 lead	83 Bi 209.0 bismuth	84 Po [209] polonium	85 At [210] astatine	86 Rn [222] radon
7	87 Fr [223] francium	88 Ra [226] radium	89-103 Ac–Lr **	104 Rf [267] rutherfordium	105 Db [270] dubnium	106 Sg [271] seaborgium	107 Bh [270] bohrium	108 Hs [277] hassium	109 Mt [276] meitnerium	110 Ds [281] darmstadtium	111 Rg [282] roentgenium	112 Cn [285] copernicium	113 Nh [285] nihonium	114 Fl [289] flerovium	115 Mc [288] moscovium	116 Lv [293] livermorium	117 Ts [294] tennessine	118 Og [294] oganesson

*	57 La 138.9 lanthanum	58 Ce 140.1 cerium	59 Pr 140.9 praseodymium	60 Nd 144.2 neodymium	61 Pm [145] promethium	62 Sm 150.4 samarium	63 Eu 152.0 europium	64 Gd 157.3 gadolinium	65 Tb 158.9 terbium	66 Dy 162.5 dysprosium	67 Ho 164.9 holmium	68 Er 167.3 erbium	69 Tm 168.9 thulium	70 Yb 173.1 ytterbium	71 Lu 175.0 lutetium
**	89 Ac [227] actinium	90 Th 232.0 thorium	91 Pa 231.0 protactinium	92 U 238.0 uranium	93 Np [237] neptunium	94 Pu [244] plutonium	95 Am [243] americium	96 Cm [247] curium	97 Bk [247] berkelium	98 Cf [251] californium	99 Es [252] einsteinium	100 Fm [257] fermium	101 Md [258] mendelevium	102 No [259] nobelium	103 Lr [262] lawrencium



Color Code	
Metal	Solid
Metalloid	Liquid
Nonmetal	Gas